

Characteristics that Determine Dungeons and Dragons Monsters' Hitpoints

Dylan Smith

2026-04-03

Abstract

This project analyzes the characteristics of Dungeons and Dragons 5th Edition monsters with the goal of predicting their hit points. Multiple linear regression was used with hit points as the response variable and size, armor class, legendary status, strength, dexterity, constitution, intelligence, wisdom, charisma, and proficiency bonus used as the initial predictor variables.

Section 1. Data Introduction and Description

Introduction

The main question this project is attempting to answer is: which Dungeons and Dragons 5th Edition monster characteristics most explain and predict their hit points? The response variable is hp, which measures a monster's hit points. The predictor variables considered in this project are challenge rating (cr), size (size), armor class (ac), legendary status (legendary), strength (str), dexterity (dex), constitution (con), intelligence (int), wisdom (wis), charisma (cha), and proficiency bonus (proficiency_bonus).

First, descriptive statistics, histograms, and boxplots are used to summarize the data. Next, multicollinearity is checked using the Variance Inflation Factor (VIF). Regression assumptions are evaluated using residual plots, the Durbin-Watson test, Q-Q plots, and the Shapiro-Wilk test. Due to the original model violating several of the regression assumptions, a Box-Cox transformation is conducted on the response variable. After refitting the model with the transformed response variable, hypothesis tests are conducted for the overall model and the predictor variables. Stepwise selection using AIC is used to discard ineffective predictor variables to obtain a more refined final model. Finally, leverage values are compared between the initial and the final models.

Descriptive Statistics and Graphics: Quantitative Variables

Table 1: Descriptive Statistics for HP, STR, and INT

Variable	Min	Q1	Median	Q3	Max	Mean	Variance	SD
hp	1	19	45	112	676	81.463840	9617.67931	98.069768
str	1	11	16	19	30	15.231920	42.15858	6.492964
int	1	3	8	12	26	8.324189	32.42464	5.694264

Figure 1: Histogram of Hit Points (HP)

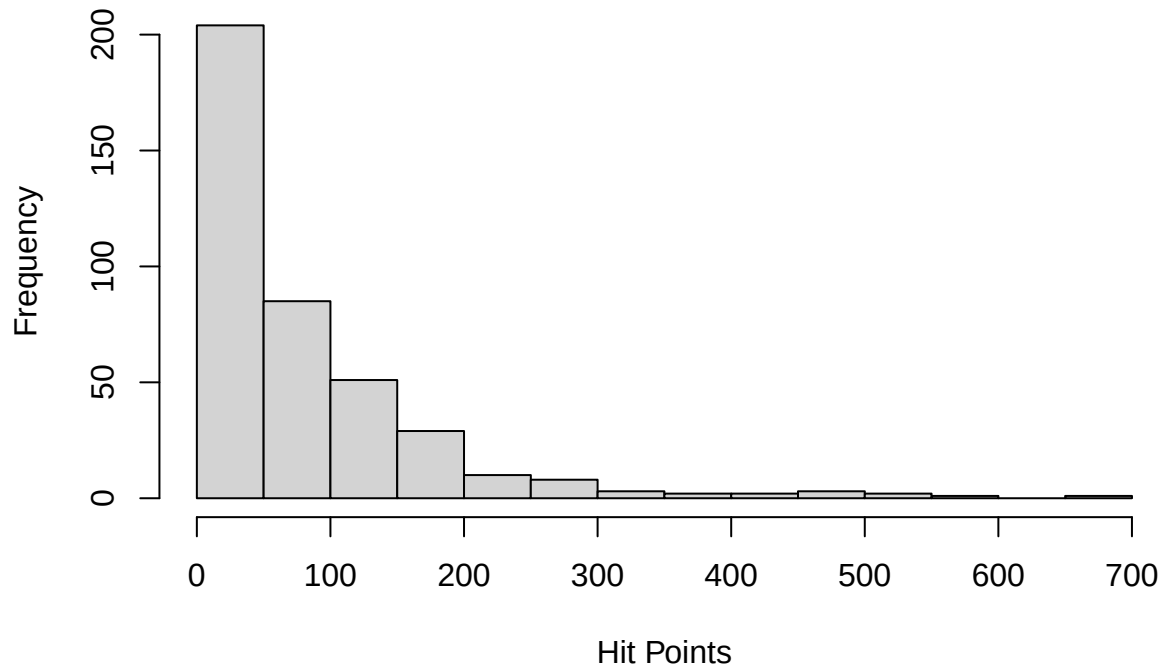
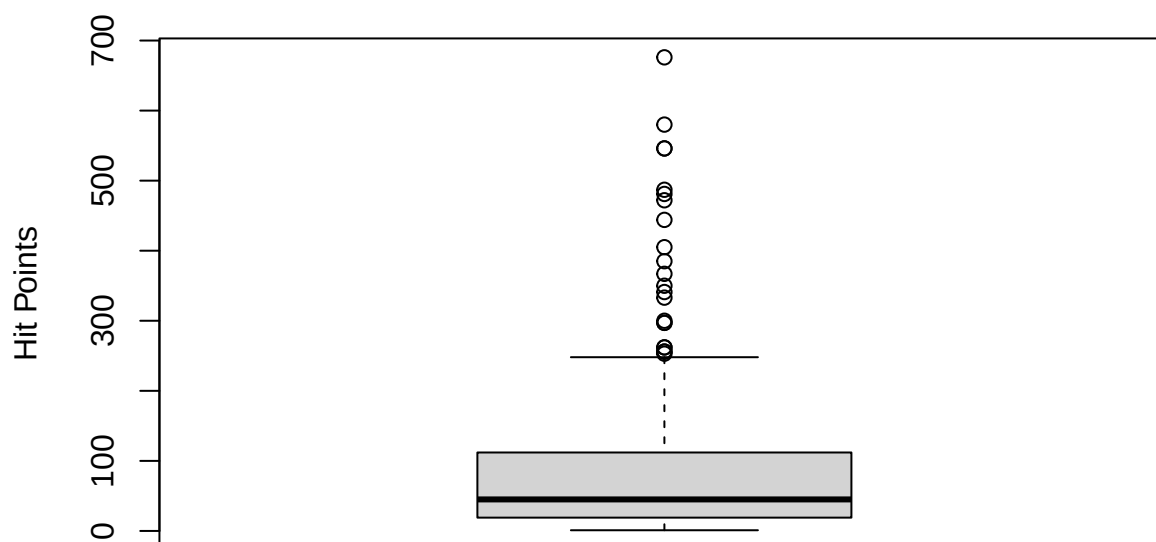


Figure 2: Boxplot of Hit Points (HP)



The histogram of HP shown in Figure 1 is strongly skewed right. This means that the monsters' hit points are not normally distributed. Most monsters have relatively low HP values. The boxplot shown in Figure 2 shows many outliers of monsters with high HP values between 300 and 700. It is possible that these outliers with high HP values correspond to boss monsters whereas the vast majority of regular monsters have much lower HP values.

Figure 3: Histogram of Strength (STR)

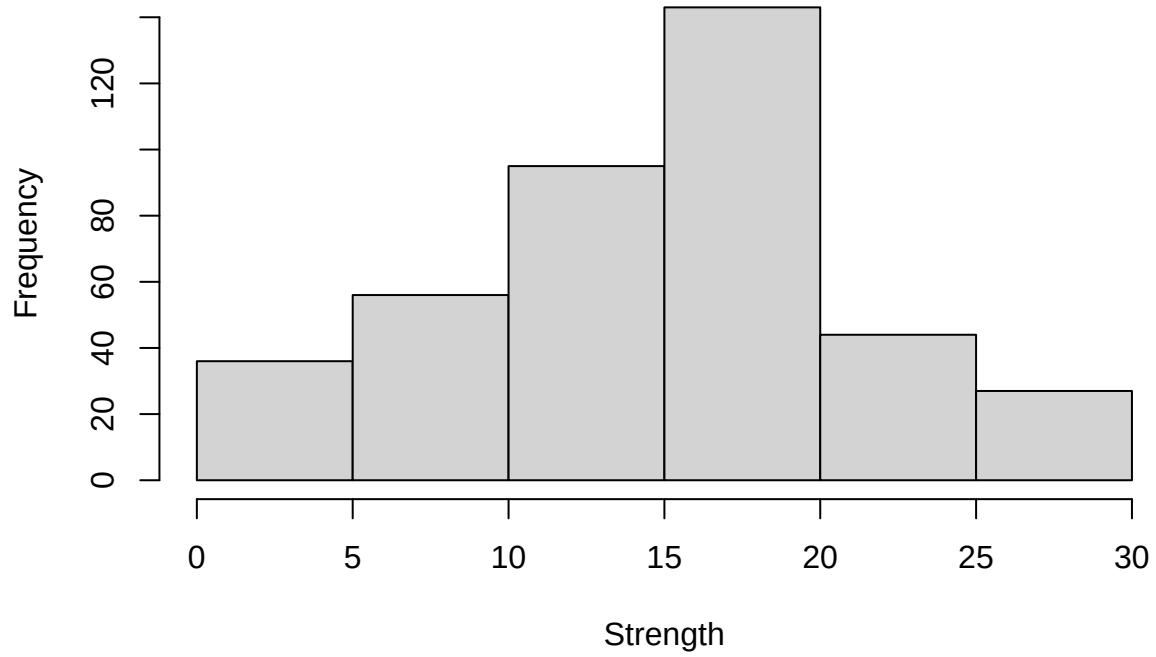
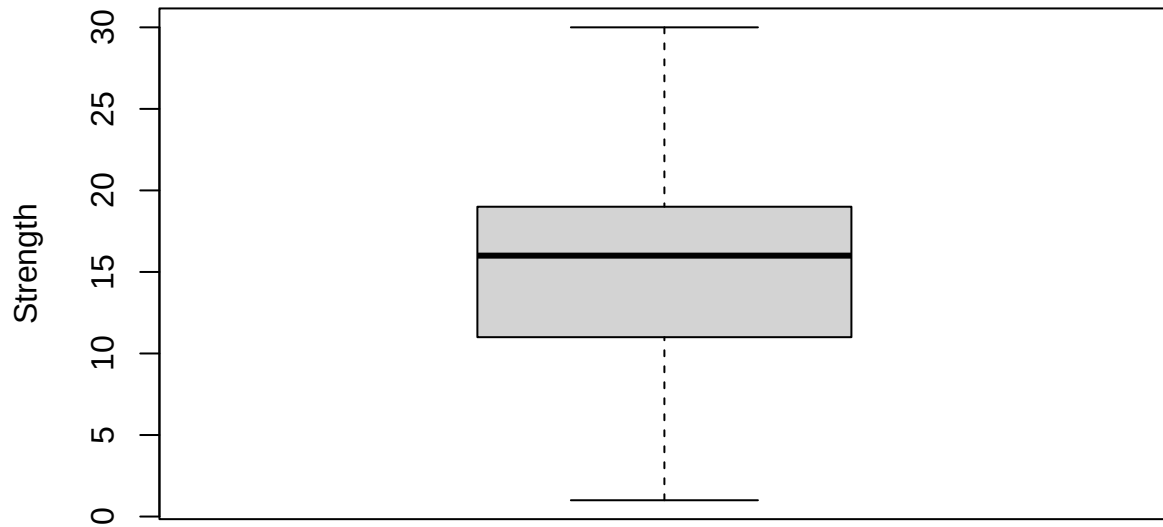


Figure 4: Boxplot of Strength (STR)



The histogram of Strength shown in Figure 3 is shaped more like a bell curve, indicating that the monsters' strength values are more normally distributed. Although the histogram is not perfectly symmetric, there is no strong skew to either side. This is further reinforced upon looking at the boxplot in Figure 4, which shows no apparent outliers present.

Figure 5: Histogram of Intelligence (INT)

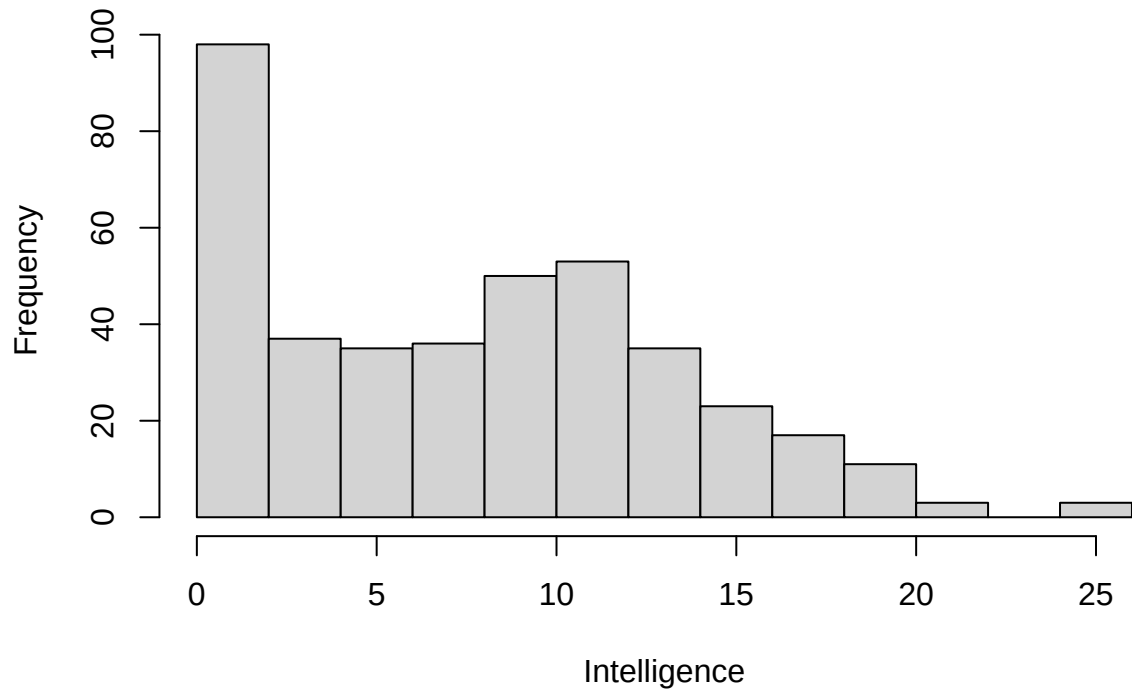
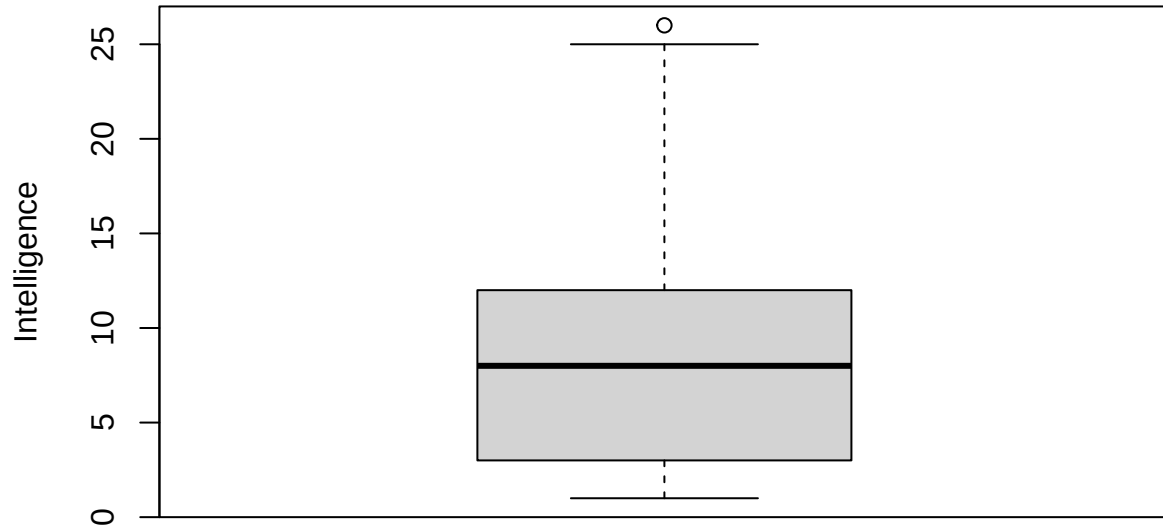


Figure 6: Boxplot of Intelligence (INT)



The histogram of Intelligence shown in Figure 5 is skewed right. This means that most monsters have low intelligence values, with the highest frequencies near the lower end and the distribution trailing off gradually as intelligence increases. The boxplot in Figure 6 shows only one outlier with an intelligence value greater than 25.

Descriptive Statistics: Indicator Variables

Table 2: Frequency Table for Creature Size

	Size	Frequency
0	Small	222
1	Large	179

The frequency table for the variable size shown in Table 2 shows that there are 222 small creatures and 179 large creatures in the data set.

Table 3: Frequency Table for Legendary Status

	Legendary	Frequency
0	Not Legendary	366
1	Legendary	35

The frequency table for the variable legendary shown in Table 3 shows that there are 366 not legendary creatures and 35 legendary creatures in the data set.

Section 2. Multiple Linear Regression Analysis

Throughout this section, a significance level of $\alpha = 0.03$ will be used to make decisions.

Multicollinearity Check

To check for multicollinearity, we use the VIF (Variance Inflation Factor) technique. For each predictor variable, VIF measures how much the variance of its estimated regression coefficient is inflated due to linear relationships with the other predictor variables. If the $VIF > 10$, that indicates that there is problematic multicollinearity present. Starting with the predictor variables of cr, size, ac, legendary, str, dex, con, int, wis, cha, and proficiency_bonus, the VIF is calculated for each predictor variable and the predictor variable with the largest VIF that exceeds 10 is removed and the model is then refitted. This process is repeated until all remaining predictor variables have VIF values less than or equal to 10.

Iteration 1:

Table 4: VIF Values for Iteration 1

Predictor	VIF
cr	37.285970
size	2.125204
ac	2.919221
legendary	2.424654
str	4.552684
dex	1.607907
con	6.817118
int	5.813308
wis	2.470425
cha	7.604695
proficiency_bonus	29.540485

Predictor variable removed: cr

Iteration 2:

Table 5: VIF Values for Iteration 2

Predictor	VIF
size	2.089099
ac	2.896041
legendary	2.421179
str	4.509599
dex	1.604906
con	6.458083
int	5.671748
wis	2.466435
cha	7.562109
proficiency_bonus	5.653765

After the second iteration, all predictor variables have VIF values less than 10. Therefore, no additional predictor variables need to be removed.

Initial Regression Model and Assumptions Check

After removing the predictor variable cr due to multicollinearity, the initial regression model was fit using hp as the response variable. The predictor variables are size, ac, legendary, str, dex, con, int, wis, cha, and proficiency_bonus. The model examines how each of these predictor variables are related to a monster's hit points.

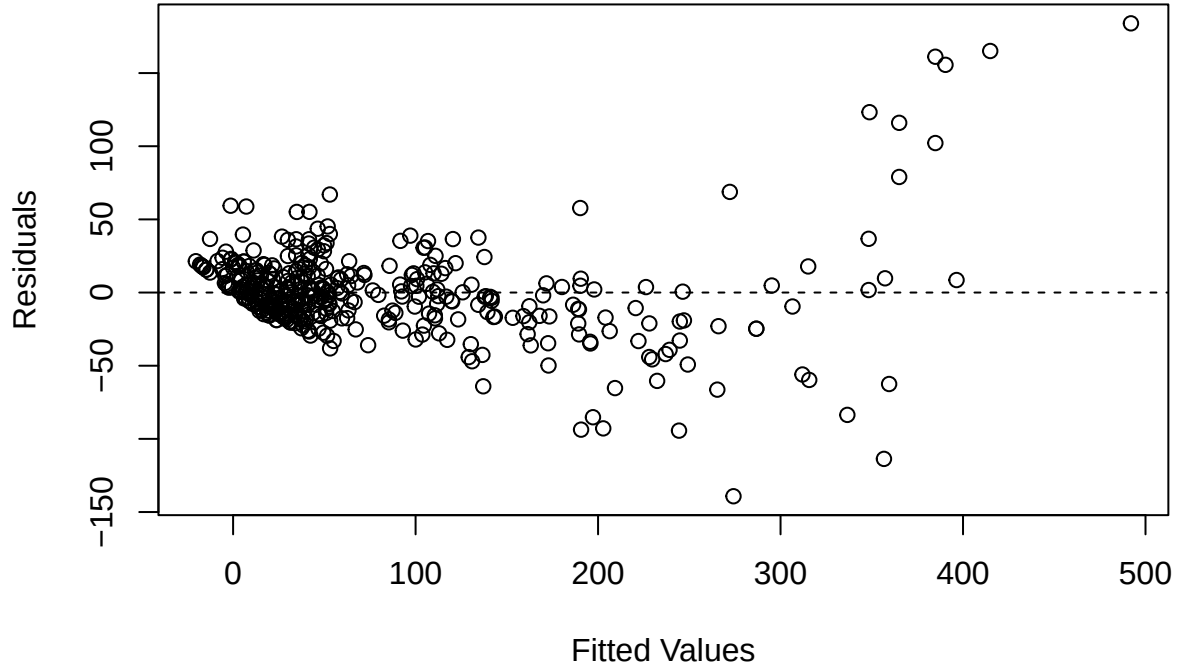
Table 6: Regression Coefficients for the Initial Model

Term	Estimate	Std. Error	t value	p-value
(Intercept)	-162.6674987	12.3366575	-13.1857027	0.0000000
size	1.0020073	4.6860857	0.2138261	0.8307944
ac	0.0378992	0.8676505	0.0436803	0.9651816
legendary	2.7382395	8.8853105	0.3081760	0.7581129
str	1.3782912	0.5277856	2.6114607	0.0093635
dex	1.1583340	0.6421574	1.8038162	0.0720316
con	6.4720349	0.9356440	6.9171982	0.0000000
int	-0.7401492	0.6749202	-1.0966470	0.2734724
wis	-1.8329278	0.8193378	-2.2370844	0.0258452
cha	0.9165342	0.7789784	1.1765849	0.2400789
proficiency_bonus	46.0060164	2.8176486	16.3278050	0.0000000

Table 7: Overall Model Summary

R-squared	Adjusted R-squared	Residual SE	F Statistic	Model p-value	df Model	df Residual
0.8944016	0.891694	32.27465	330.3238	< 0.0001	10	390

Figure 7: Residuals vs Fitted Values



The linearity check is performed using the Residuals vs Fitted Values plot shown in Figure 7. The plot shows that the residuals are not randomly distributed around the value 0. The residuals appear to be distributed in a curved pattern. Therefore, the linearity assumption is not met.

The equal variance of residuals check is also performed using the Residuals vs Fitted Values plot shown in Figure 7. Because the plot shows that the residuals are not randomly distributed around 0 and are instead distributed in a curved pattern, the equal variance of residuals assumption is also not met.

Table 8: Durbin-Watson Test for Independence of Residuals

Lag	Autocorrelation	D.W.Statistic	p.value
1	0.0939334	1.810101	0.072

The independence check is performed using the Durbin-Watson Test shown in Table 8. The seed is set to 12 to ensure that the p-value generated by the Durbin-Watson Test is reproducible.

H_0 : The residuals are independent.

H_1 : The residuals are not independent.

p-value: $p = 0.072$

Decision: Do not reject H_0 . $0.072 > 0.03$

Conclusion: There is not enough evidence to conclude that the residuals are dependent. The independence assumption is met.

Figure 8: QQ Plot of Residuals

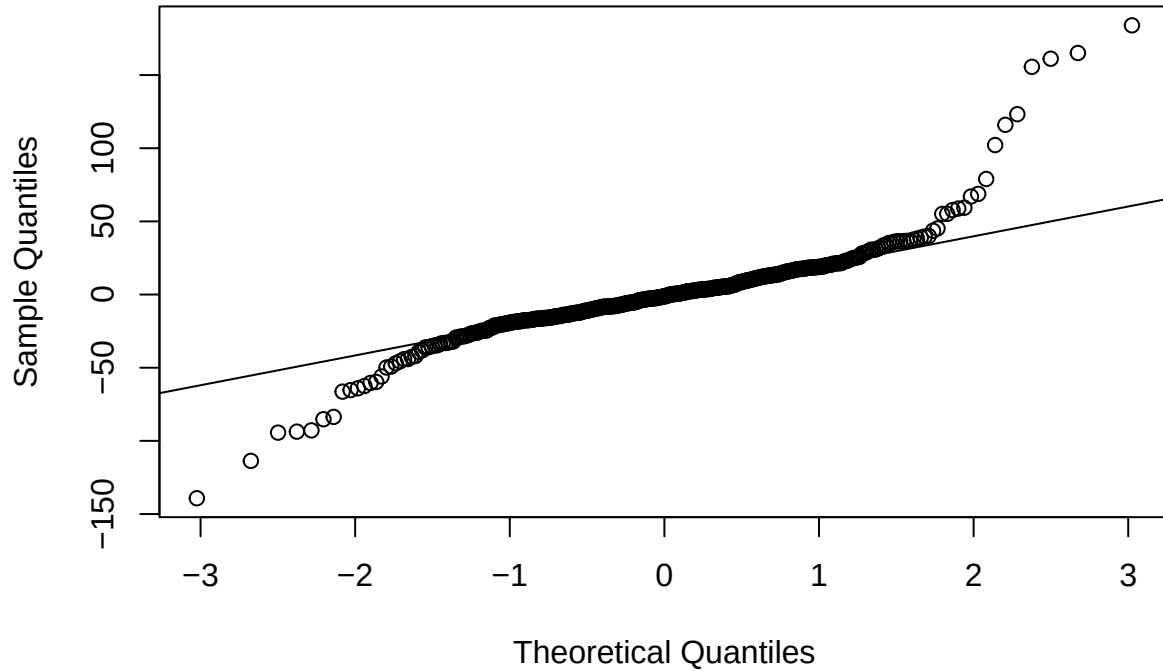


Table 9: Shapiro-Wilk Normality Test

W-Statistic	p-value
0.8486197	< 0.0001

The normality check is performed with the QQ plot shown in Figure 8. The QQ plot shows that the points deviate substantially at both tails. If the residuals are normally distributed, the points on the QQ plot should fall approximately on the reference line. Because there is so much deviation at the tails, the normality assumption is not met.

The normality check is also performed using the Shapiro-Wilk Test shown in Table 9.

H_0 : The residuals are normally distributed.

H_1 : The residuals are not normally distributed.

p-value: $p < 0.0001$

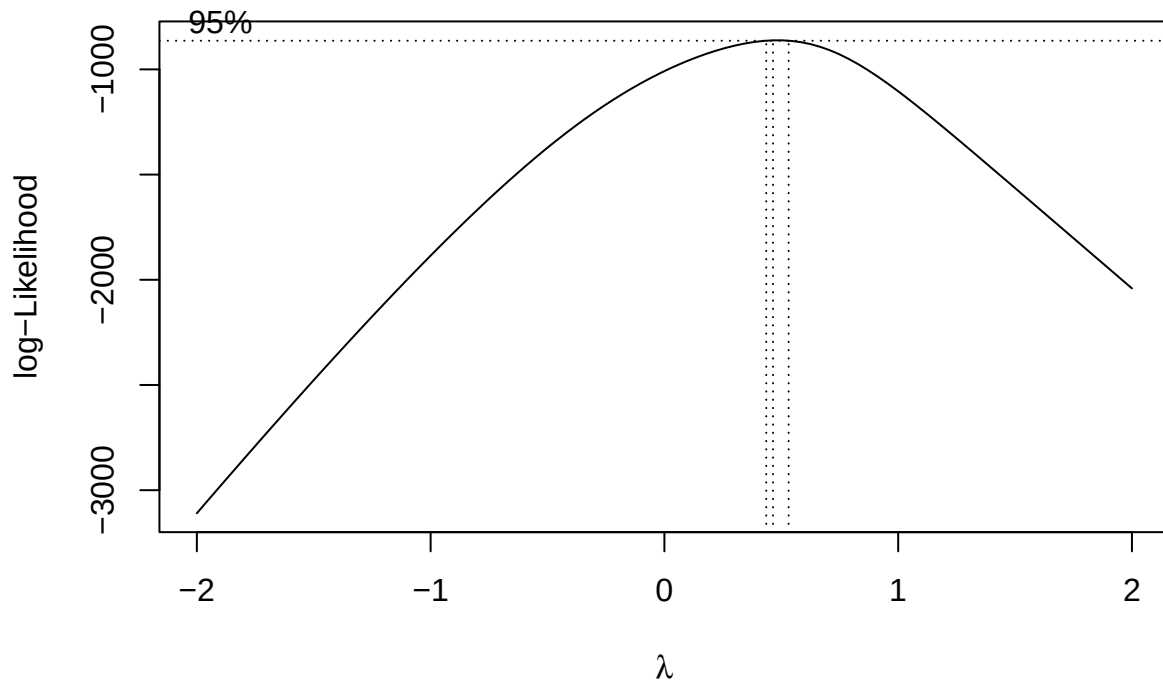
Decision: Reject H_0 . $p < 0.03$

Conclusion: There is enough evidence to conclude that the residuals are not normally distributed. The normality assumption is not met.

Transformation #1 (Y-Variable)

The linearity, equal variance, and normality assumptions were all not met. Based on this, it does seem necessary to transform the original hp variable. This is necessary because the current model is suggesting that the hp variable is not fitting well.

Figure 9: Box-Cox Transformation

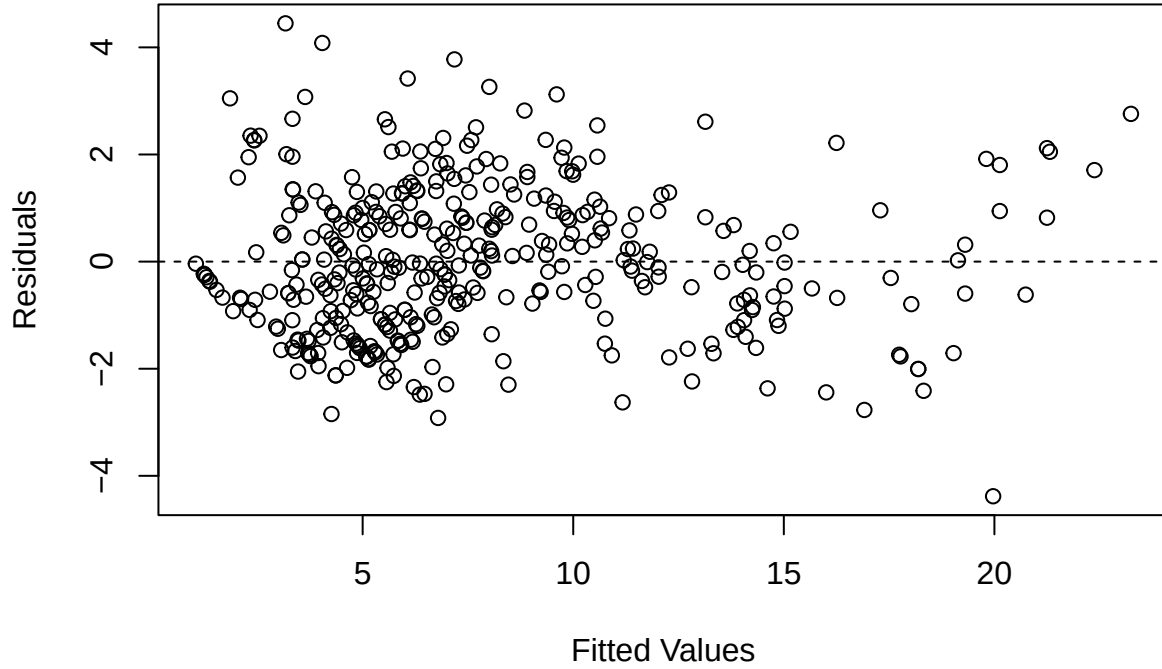


The Box-Cox plot shown in Figure 9 suggests that a value of $\lambda = 0.5$ is best. A new column has been added to the dataset corresponding to $hp^{0.5}$

Regression Model and Assumptions Check after Transformation #1

After applying the Box-Cox transformation, a new model is needed with the transformed response variable. The predictor variables remain unchanged. The predictor variables are size, ac, legendary, str, dex, con, int, wis, cha, and proficiency_bonus.

Figure 10: Residuals vs Fitted Values



The linearity check is performed using the Residuals vs Fitted Values plot shown in Figure 10. The plot shows that the residuals are now randomly distributed around the value 0. Therefore, the linearity assumption is met.

The equal variance of residuals check is also performed using the Residuals vs Fitted Values plot shown in Figure 10. Because the plot shows that the residuals are randomly distributed around 0, the equal variance of residuals assumption is also met.

Table 10: Durbin-Watson Test for Independence of Residuals

Lag	Autocorrelation	D.W.Statistic	p.value
1	0.1293421	1.738227	0.012

The independence check is performed using the Durbin-Watson Test shown in Table 10. The seed is set to 12 to ensure that the p-value generated by the Durbin-Watson Test is reproducible.

H_0 : The residuals are independent.

H_1 : The residuals are not independent.

p-value: $p = 0.012$

Decision: Reject H_0 . $0.012 < 0.03$

Conclusion: There is enough evidence to conclude that the residuals are not independent. The independence assumption is not met.

Figure 11: QQ Plot of Residuals

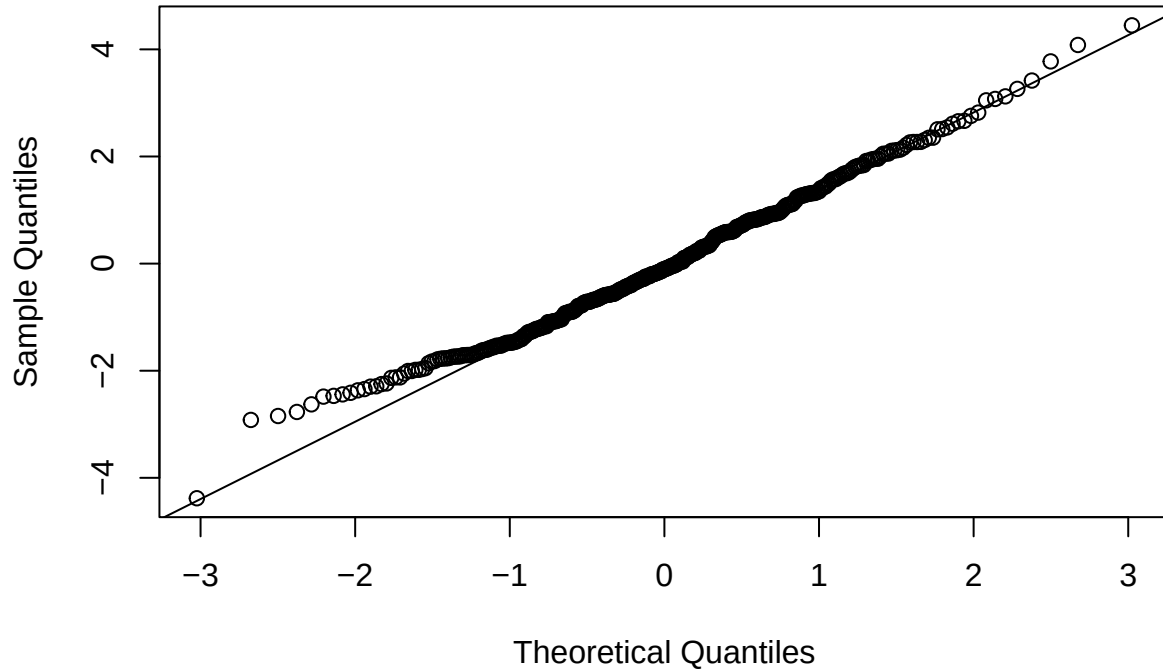


Table 11: Shapiro-Wilk Normality Test

W-Statistic	p-value
0.9921441	0.0328136

The normality check is performed with the QQ plot shown in Figure 11. The QQ plot shows that the points deviate substantially less at both tails than they did before. The points on the QQ plot now lie much closer to the reference line. Because of this, the normality assumption is met.

The normality check is also performed using the Shapiro-Wilk Test shown in Table 11.

H_0 : The residuals are normally distributed.

H_1 : The residuals are not normally distributed.

p-value: $p = 0.0328136$

Decision: Do not reject H_0 . $0.0328136 > 0.03$

Conclusion: There is enough evidence to conclude that the residuals are normally distributed. The normality assumption is met.

Transformation #2 (X-Variable)

No additional X-Variable transformation is needed. The linearity, equal variance, and normality assumptions, which were previously not met before the Y-Variable transformation, are now met. The only issue is that now the independence assumption is no longer met at a significance level of 0.03. However, as X-Variable transformation is unlikely to fix this problem and result in all four checks being met.

Hypothesis Testing

Table 12: Regression Coefficients for the Transformed Model

Term	Estimate	Std. Error	t value	p-value
(Intercept)	-4.9094882	0.5290920	-9.2790832	0.0000000
size	0.6751407	0.2009758	3.3593128	0.0008584
ac	-0.0057771	0.0372116	-0.1552510	0.8767038
legendary	-1.4381654	0.3810713	-3.7740060	0.0001856
str	0.1280031	0.0226356	5.6549562	0.0000000
dex	0.0334853	0.0275407	1.2158482	0.2247782
con	0.3988842	0.0401277	9.9403706	0.0000000
int	0.0975140	0.0289458	3.3688436	0.0008300
wis	-0.0789030	0.0351396	-2.2454172	0.0253009
cha	0.0183713	0.0334087	0.5498966	0.5827050
proficiency_bonus	1.4730170	0.1208427	12.1895388	0.0000000

Table 13: Overall Model Summary

R-squared	Adjusted R-squared	Residual SE	F Statistic	Model p-value	df Model	df Residual
0.912529	0.9102861	1.384188	406.8618	< 0.0001	10	390

Overall Model Test:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_{10} = 0$$

$$H_1 : \text{At least one } \beta_j \neq 0$$

$$F = 406.8618$$

$$p < 0.0001$$

Decision: Reject H_0 . $p < 0.0001 < 0.03$.

Conclusion: There is sufficient evidence to conclude that at least one predictor variable is linearly related to the transformed response variable for hit points.

Individual Predictor Variable Tests:

Size:

$$H_0 : \beta_{\text{size}} = 0$$

$$H_1 : \beta_{\text{size}} \neq 0$$

$$t = 3.3593128 \quad p = 0.0008584$$

Decision: Reject H_0 . $p = 0.0008584 < 0.03$ Conclusion: There is enough evidence to conclude that size is a significant predictor variable for the transformed response variable for hit points.

Armor Class:

$$H_0 : \beta_{\text{ac}} = 0$$

$$H_1 : \beta_{\text{ac}} \neq 0$$

$$t = -0.1552510 \quad p = 0.8767038$$

Decision: Do not reject H_0 . $p = 0.8767038 > 0.03$ Conclusion: There is not enough evidence to conclude that armor class is a significant predictor variable for the transformed response variable for hit points.

Legendary:

$$H_0 : \beta_{\text{legendary}} = 0$$

$$H_1 : \beta_{\text{legendary}} \neq 0$$

$$t = -3.7740060 \quad p = 0.0001856$$

Decision: Reject H_0 . $p = 0.0001856 < 0.03$ Conclusion: There is enough evidence to conclude that legendary is a significant predictor variable for the transformed response variable for hit points.

Strength:

$$H_0 : \beta_{\text{str}} = 0$$

$$H_1 : \beta_{\text{str}} \neq 0$$

$$t = 5.6549562 \quad p < 0.0001$$

Decision: Reject H_0 . $p < 0.0001 < 0.03$ Conclusion: There is enough evidence to conclude that strength is a significant predictor variable for the transformed response variable for hit points.

Dexterity:

$$H_0 : \beta_{\text{dex}} = 0$$

$$H_1 : \beta_{\text{dex}} \neq 0$$

$$t = 1.2158482 \quad p = 0.2247782$$

Decision: Do not reject H_0 . $p = 0.2247782 > 0.03$ Conclusion: There is not enough evidence to conclude that dexterity is a significant predictor variable for the transformed response variable for hit points.

Constitution:

$$H_0 : \beta_{\text{con}} = 0$$

$$H_1 : \beta_{\text{con}} \neq 0$$

$$t = 9.9403706 \quad p < 0.0001$$

Decision: Reject H_0 . $p < 0.0001 < 0.03$ Conclusion: There is enough evidence to conclude that constitution is a significant predictor variable for the transformed response variable for hit points.

Intelligence:

$$H_0 : \beta_{\text{int}} = 0$$

$$H_1 : \beta_{\text{int}} \neq 0$$

$$t = 3.3688436 \quad p = 0.0008300$$

Decision: Reject H_0 . $p = 0.0008300 < 0.03$ Conclusion: There is enough evidence to conclude that intelligence is a significant predictor variable for the transformed response variable for hit points.

Wisdom:

$$H_0 : \beta_{\text{wis}} = 0$$

$$H_1 : \beta_{\text{wis}} \neq 0$$

$$t = -2.2454172 \quad p = 0.0253009$$

Decision: Reject H_0 . $p = 0.0253009 < 0.03$ Conclusion: There is enough evidence to conclude that wisdom is a significant predictor variable for the transformed response variable for hit points.

Charisma:

$$H_0 : \beta_{\text{cha}} = 0$$

$$H_1 : \beta_{\text{cha}} \neq 0$$

$$t = 0.5498966 \quad p = 0.5827050$$

Decision: Do not reject H_0 . $p = 0.5827050 > 0.03$ Conclusion: There is not enough evidence to conclude that charisma is a significant predictor variable for the transformed response variable for hit points.

Proficiency Bonus:

$$H_0 : \beta_{\text{proficiency_bonus}} = 0$$

$$H_1 : \beta_{\text{proficiency_bonus}} \neq 0$$

$$t = 12.1895388 \quad p < 0.0001$$

Decision: Reject H_0 . $p < 0.0001 < 0.03$ Conclusion: There is enough evidence to conclude that proficiency bonus is a significant predictor variable for the transformed response variable for hit points.

R² Value: 0.912529

An R^2 Value of $R^2 = 0.912529$ means that approximately 91.2529% of the variability in the transformed response variable hit points is explained by the predictors in the model.

Section 3. Variable Selection

To build the “best” model possible, stepwise selection using AIC was used. This procedure combines elements of forward selection and backward elimination. Variables are added to the model sequentially. After adding each new variable, variables that no longer provide an improvement to best model fit are removed.

Table 14: Regression Coefficients for the Best Model

Term	Estimate	Std. Error	t value	p-value
(Intercept)	-4.6409365	0.4067116	-11.410877	0.0000000
size	0.6724299	0.1987886	3.382638	0.0007900
legendary	-1.4867110	0.3732720	-3.982916	0.0000811
str	0.1258033	0.0219168	5.740049	0.0000000
con	0.3920857	0.0387619	10.115236	0.0000000
int	0.1121865	0.0179358	6.254894	0.0000000
wis	-0.0577261	0.0311126	-1.855395	0.0642884
proficiency_bonus	1.4844378	0.1167570	12.713907	0.0000000

Table 15: Overall Model Summary

R-squared	Adjusted R-squared	Residual SE	F Statistic	Model p-value	df Model	df Residual
0.9120836	0.9105176	1.382401	582.4505	< 0.0001	7	393

After the stepwise selection procedure, the variables we are left with are size, legendary, str, con, int, wis, and proficiency_bonus. The variables ac, dex, and cha are removed. The best value has an R^2 value of $R^2 = 0.9120836$. This means that approximately 91.20836% of variability in the transformed response variable hit points can be explained by the predictor variables size, legendary, str, con, int, wis, and proficiency_bonus.

Section 4. High Leverage Point Comparison

Table 16: Leverage Cutoffs for the Two Models

Model	Predictors	n	Average_Leverage	Cutoff
Initial Model	10	401	0.0274314	0.0822943
Best Model	7	401	0.0199501	0.0598504

Table 17: High Leverage Points in the Initial Model

Observation	Leverage
35	0.0866982
225	0.0914681
317	0.1983846
344	0.1322252
377	0.0865018
400	0.0984770

Table 18: High Leverage Points in the Best Model

Observation	Leverage
2	0.0647352
103	0.0605130
108	0.0634008
222	0.0689827
225	0.0776725
317	0.1122674
344	0.1223241
355	0.0724109
400	0.0710469

There are high leverage points present in both models. For the initial model, the cutoff is 0.0822943. The observations with high leverage in the base model are observations 35, 225, 317, 344, 377, and 400. In the best model, the cutoff is 0.0598504. The observations with high leverage in the best model are observations 2, 103, 108, 222, 225, 317, 344, 355, and 400. Observations 35 and 377 are high leverage only in the base model. These observations changed from high leverage to “normal” after the variable selection in the best model. Observations 2, 103, 108, 222, and 355 are high leverage only in the best model. Observations 225, 317, 344, and 400 are high leverage in both models.

Section 5. Conclusions

Final Statistical Model: $\hat{Y} = -4.6409365 + 0.6724299x_1 - 1.4867110x_2 + 0.1258033x_3 + 0.3920857x_4 + 0.1121865x_5 - 0.0577261x_6 + 1.4844378x_7$

$\hat{Y} = \sqrt{hp}$

x_1 = size

x_2 = legendary

x_3 = str

x_4 = con

x_5 = int

x_6 = wis

x_7 = proficiency_bonus

R² Comparison:

Before variable selection, the model had an R^2 value of $R^2 = 0.912529$. After variable selection, the best model had an R^2 value of $R^2 = 0.9120836$. The R^2 value decreased slightly, but still remained relatively the same. This indicates that the original model explained nearly the same amount of variability in hit points as the final model.

Interpretation of One Variable

Using the variable strength (str), as a monster’s strength increases by one point, the $\sqrt{hp}$ of the monster increases by 0.1258033.

Appendix

Code

```
library(knitr)

data <- read.csv("DnD_5e_Monsters_481.csv", header = TRUE)
vars <- c("hp", "str", "int")

descriptive_stats <- function(x) {
```

```

c(
  Min = min(x, na.rm = TRUE),
  Q1 = quantile(x, 0.25, na.rm = TRUE),
  Median = median(x, na.rm = TRUE),
  Q3 = quantile(x, 0.75, na.rm = TRUE),
  Max = max(x, na.rm = TRUE),
  Mean = mean(x, na.rm = TRUE),
  Variance = var(x, na.rm = TRUE),
  SD = sd(x, na.rm = TRUE)
)
}

summary_table <- data.frame(
  Variable = vars,
  t(sapply(data[vars], descriptive_stats)),
  row.names = NULL
)

colnames(summary_table) <- c("Variable", "Min", "Q1", "Median", "Q3", "Max", "Mean", "Variance", "SD")

kable(summary_table, caption = "Descriptive Statistics for HP, STR, and INT")

hist(data$hp, main = "Figure 1: Histogram of Hit Points (HP)", xlab = "Hit Points", ylab = "Frequency")
boxplot(data$hp, main = "Figure 2: Boxplot of Hit Points (HP)", ylab = "Hit Points")

hist(data$str, main = "Figure 3: Histogram of Strength (STR)", xlab = "Strength", ylab = "Frequency")
boxplot(data$str, main = "Figure 4: Boxplot of Strength (STR)", ylab = "Strength")

hist(data$int, main = "Figure 5: Histogram of Intelligence (INT)", xlab = "Intelligence", ylab = "Frequency")
boxplot(data$int, main = "Figure 6: Boxplot of Intelligence (INT)", ylab = "Intelligence")

size_table <- table(data$size)

size_summary <- data.frame(
  Size = c("Small", "Large"),
  Frequency = c(size_table["0"], size_table["1"])
)

kable(size_summary, caption = "Frequency Table for Creature Size")

legendary_table <- table(data$legendary)

legendary_summary <- data.frame(
  Legendary = c("Not Legendary", "Legendary"),
  Frequency = c(legendary_table["0"], legendary_table["1"])
)

kable(legendary_summary, caption = "Frequency Table for Legendary Status")

alpha <- 0.03
model <- lm(hp ~ cr + size + ac + legendary + str + dex + con + int + wis + cha + proficiency_bonus, data = data)

library(car)

```

```

vif1 <- vif(model)

vif1_table <- data.frame(
  Predictor = names(vif1),
  VIF = as.numeric(vif1)
)

kable(vif1_table, caption = "VIF Values for Iteration 1")

model2 <- lm(hp ~ size + ac + legendary + str + dex + con + int + wis + cha + proficiency_bonus, data =

vif2 <- vif(model2)

vif2_table <- data.frame(
  Predictor = names(vif2),
  VIF = as.numeric(vif2)
)

kable(vif2_table, caption = "VIF Values for Iteration 2")

library(broom)
coefficient_table <- tidy(model2)[, c("term", "estimate", "std.error", "statistic", "p.value")]

kable(coefficient_table, col.names = c("Term", "Estimate", "Std. Error", "t value", "p-value"), caption

fit_table <- glance(model2)[, c("r.squared", "adj.r.squared", "sigma", "statistic", "p.value", "df", "d
fit_table$p.value <- ifelse(fit_table$p.value < 0.0001, "< 0.0001", fit_table$p.value)

kable(fit_table, col.names = c("R-squared", "Adjusted R-squared", "Residual SE", "F Statistic", "Model p

plot(fitted(model2), residuals(model2),
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Figure 7: Residuals vs Fitted Values")
abline(h = 0, lty = 2)

set.seed(12)
dw <- durbinWatsonTest(model2)

dw_table <- data.frame(
  Lag = 1,
  Autocorrelation = dw$r,
  'D-W Statistic' = dw$dw,
  'p-value' = dw$p,
  row.names = NULL
)

kable(dw_table, caption = "Durbin-Watson Test for Independence of Residuals")

qqnorm(residuals(model2), main = "Figure 8: QQ Plot of Residuals")
qqline(residuals(model2))

shapiro <- shapiro.test(residuals(model2))

```

```

shapiro_table <- data.frame(
  W = shapiro$statistic,
  p_value = ifelse(shapiro$p.value < 0.0001, "< 0.0001", shapiro$p.value),
  row.names = NULL
)

kable(shapiro_table, col.names = c("W-Statistic", "p-value"), caption = "Shapiro-Wilk Normality Test")

library(MASS)
boxcox(model2)
title(main = "Figure 9: Box-Cox Transformation")

data$hp_transformed <- sqrt(data$hp)

model3 <- lm(hp_transformed ~ size + ac + legendary + str + dex + con + int + wis + cha + proficiency_b

plot(fitted(model3), residuals(model3),
     xlab = "Fitted Values",
     ylab = "Residuals",
     main = "Figure 10: Residuals vs Fitted Values")
abline(h = 0, lty = 2)

set.seed(12)
dw <- durbinWatsonTest(model3)

dw_table <- data.frame(
  Lag = 1,
  Autocorrelation = dw$r,
  'D-W Statistic' = dw$dw,
  'p-value' = dw$p,
  row.names = NULL
)

kable(dw_table, caption = "Durbin-Watson Test for Independence of Residuals")

qqnorm(residuals(model3), main = "Figure 11: QQ Plot of Residuals")
qqline(residuals(model3))

shapiro <- shapiro.test(residuals(model3))

shapiro_table <- data.frame(
  W = shapiro$statistic,
  p_value = ifelse(shapiro$p.value < 0.0001, "< 0.0001", shapiro$p.value),
  row.names = NULL
)

kable(shapiro_table, col.names = c("W-Statistic", "p-value"), caption = "Shapiro-Wilk Normality Test")

coefficient_table <- tidy(model3)[, c("term", "estimate", "std.error", "statistic", "p.value")]

kable(coefficient_table, col.names = c("Term", "Estimate", "Std. Error", "t value", "p-value"), caption

fit_table <- glance(model3)[, c("r.squared", "adj.r.squared", "sigma", "statistic", "p.value", "df", "d

```

```

fit_table$p.value <- ifelse(fit_table$p.value < 0.0001, "< 0.0001", fit_table$p.value)

kable(fit_table, col.names = c("R-squared", "Adjusted R-squared", "Residual SE", "F Statistic", "Model p-value"))

best_model <- stepAIC(model3, direction = "both", trace = FALSE)

coefficient_table <- tidy(best_model)[, c("term", "estimate", "std.error", "statistic", "p.value")]

kable(coefficient_table, col.names = c("Term", "Estimate", "Std. Error", "t value", "p-value"), caption = "Coefficients")

fit_table <- glance(best_model)[, c("r.squared", "adj.r.squared", "sigma", "statistic", "p.value", "df")]
fit_table$p.value <- ifelse(fit_table$p.value < 0.0001, "< 0.0001", fit_table$p.value)

kable(fit_table, col.names = c("R-squared", "Adjusted R-squared", "Residual SE", "F Statistic", "Model p-value"))

leverage_model2 <- hatvalues(model2)
leverage_best_model <- hatvalues(best_model)

n <- nrow(data)

p_model2 <- length(coef(model2)) - 1
p_best <- length(coef(best_model)) - 1

hbar_model2 <- (p_model2 + 1) / n
hbar_best <- (p_best + 1) / n

cutoff_model2 <- 3 * hbar_model2
cutoff_best <- 3 * hbar_best

high_leverage_model2 <- which(leverage_model2 > cutoff_model2)
high_leverage_best <- which(leverage_best_model > cutoff_best)

lev_summary <- data.frame(
  Model = c("Initial Model", "Best Model"),
  Predictors = c(p_model2, p_best),
  n = c(n, n),
  Average_Leverage = c(hbar_model2, hbar_best),
  Cutoff = c(cutoff_model2, cutoff_best)
)

kable(lev_summary, caption = "Leverage Cutoffs for the Two Models")

high_leverage_table1 <- data.frame(
  Observation = high_leverage_model2,
  Leverage = leverage_model2[high_leverage_model2],
  row.names = NULL
)

high_leverage_table2 <- data.frame(
  Observation = high_leverage_best,
  Leverage = leverage_best_model[high_leverage_best],
  row.names = NULL
)

```

```
)
```

```
kable(high_leverage_table1, caption = "High Leverage Points in the Initial Model")  
kable(high_leverage_table2, caption = "High Leverage Points in the Best Model")
```

Division of Work

I worked independently on this project.